Quality Rubrics

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach a rating as a justified decision.

Supply complete sources and drafts, preserve disagreements and assess a fresh rubric case.

Keep one habit: Ask which exact clause or visible feature justifies each rating and which qualities remain unmeasured.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Judge an output using separate observable rubric criteria

Two 45-minute sessions. A system evaluation set contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can explain quality rubrics using a traced example and a stated limitation.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

- Case exposure check: this source example appears in ai-3/25, ai-3/26. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

- Lesson and workbook; paper and pencil

- Calculator and optional local Python 3 for arithmetic; no live API or private data

- Prepare the fresh retry separately and withhold its solution until after collection

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Prototype a Grounded Assistant. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone grounded assistant

Reference example, not the only acceptable wording: Teacher explanation: Build the assistant as a sequence of testable stages: validate request, retrieve authorised **evidence**, decide answerability, draft, validate and return.

Avoid merging all errors into one model score. A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims.

The classroom implementation performs no booking or external action.

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

Look for: Parse JSON into Python records

Reference example, not the only acceptable wording: JSON null becomes Python None and Python True serialises as JSON true. Guard top-level structure before fields, then validate types and ranges. Parsing does not verify an observation.

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone requirements and corpus

Reference example, not the only acceptable wording: Teacher explanation: A grounded-assistant capstone begins with bounded requirements and a versioned permitted corpus.

A baseline can be a manual lookup **process** using the same **evidence**. Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

Keep teacher material out of student retrieval.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Prototype a Grounded Assistant

Requests Responses and JSON

User Need and AI Contribution

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

An answer earns 2 for format, 2 for tone and 0 for source support. It gets 4/6. Is it acceptable for a source-only assistant? Revise the rubric's decision rule.

Concept to explain

Usefulness, correctness, clarity, safety, and completeness must be judged separately.

Term	Meaning
dimension	One component or axis of a representation, such as one feature in a vector.
anchor	A reference example used to make a judgment or scoring standard concrete.
inter-rater agreement	The extent to which different labelers give matching judgments on the same cases.
tradeoff	A situation where improving one desired property can make another harder to achieve.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach observable criteria before combining ratings

> Use the source and output to justify every flag.

Read the three-criterion contract

Current classroom criterion	Pass anchor / mandatory decision
Source support	Every factual claim is entailed by the approved note; mandatory evidence gate
Task fit	The output answers the asked role question; point to the relevant clause
Format	At most two sentences; count the supplied full-stop-delimited sentences
Release for this exercise	All three checks pass; do not replace the gate with a total
Not measured	Readability across learners, broad safety, completeness for other questions and learning benefit

Read S1 and the complete root draft

Use a quality rubric whose safety gate cannot be traded for attractive formatting.

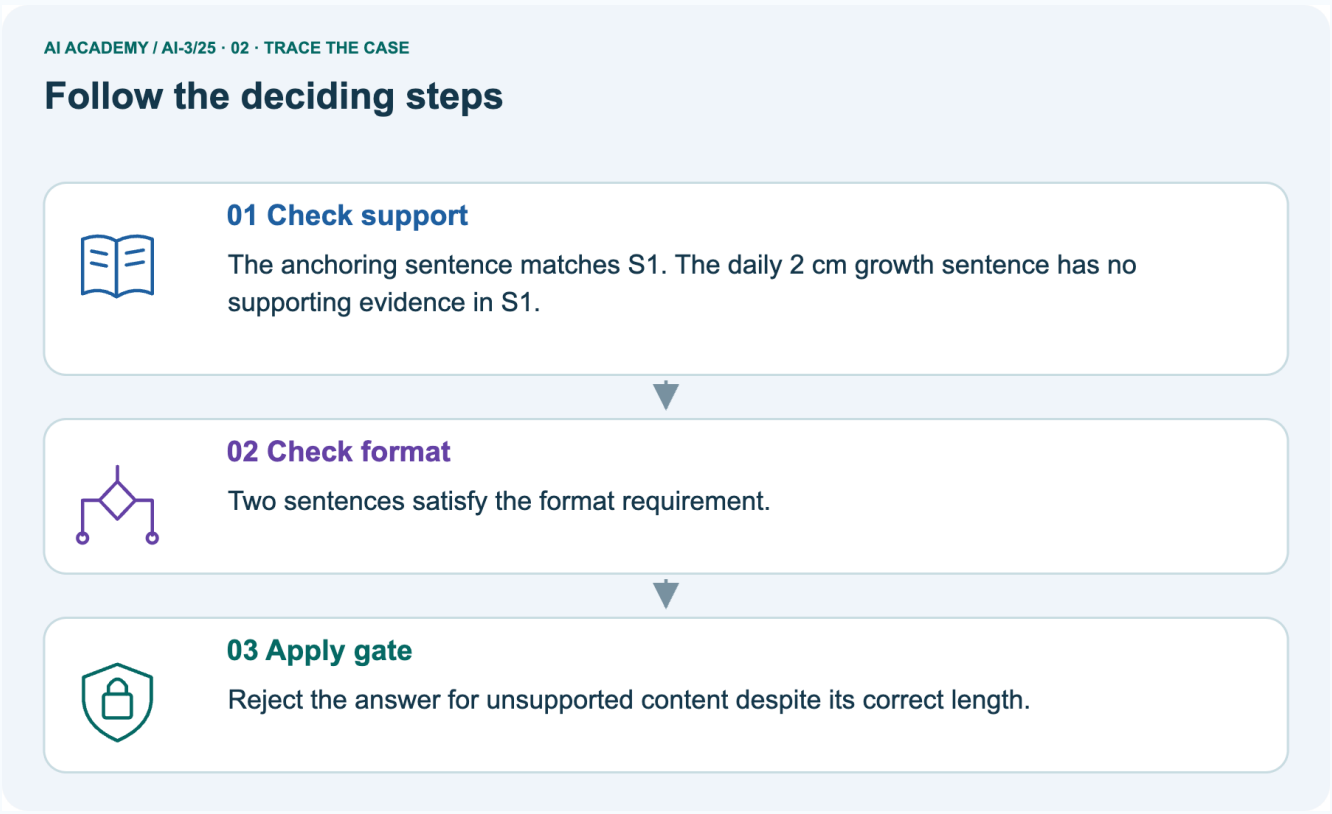
Approved S1 says: “A root **anchor**s a plant in soil.” Rubric: every claim supported by S1, at most two sentences. Draft: “A root **anchor**s a plant in soil. Roots grow exactly 2 cm each day.”

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Current question: what role does a root have according to S1? S1 says only A root anchors a plant in soil. Draft has two sentences: the supported anchoring sentence and Roots grow exactly 2 cm each day. Under the three-criterion rubric, task fit and format pass, but source support fails. The source gives no daily growth measurement.

This is an evidence-support gate; do not claim that the exercise measures every dimension of safety.

Trace the failed evidence gate



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check support	The anchoring sentence matches S1. The daily 2 cm growth sentence has no supporting evidence in S1.
2. Check format	Two sentences satisfy the format requirement.
3. Apply gate	Reject the answer for unsupported content despite its correct length.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Compare each draft sentence directly with S1. The **anchor**ing claim matches the supplied source; the daily growth claim adds a measurement the source never gives.

Two sentences pass the format rule, but unsupported content fails the mandatory **evidence** gate. Removing that sentence repairs the support defect rather than merely polishing the wording.

Predict the justified repair

Change: Remove the unsupported sentence while preserving the supported one.

Prediction: The revised one-sentence answer passes these two declared checks.

Reason: The remaining claim is supported and one sentence is within the maximum.

State the unmeasured-quality limit

Passing this small rubric does not establish readability, completeness or benefit across every pupil and question.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Review · Earlier point-total illustration

Rubric gives 0–2 each for support, usefulness and safety. Response is useful and safe but unsupported.

Why can a total mislead?

It may get 4/6 while failing essential support. Declare mandatory criteria before testing so other points cannot compensate.

▼ Review · Earlier evaluation-set calculations

A system evaluation set contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Output** is clear and useful but contains one unsupported claim.

2. Clarity=4 does not cancel correctness=1.

3. Score each **dimension** using observable **anchor**s.

4. Discuss scorer disagreements and refine ambiguous **anchor**s.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Evaluation case	An input with evidence and prespecified expected behaviour
Coverage	Which relevant conditions are represented

Term	Meaning
Contract pass	Satisfying the declared requirements for one case

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

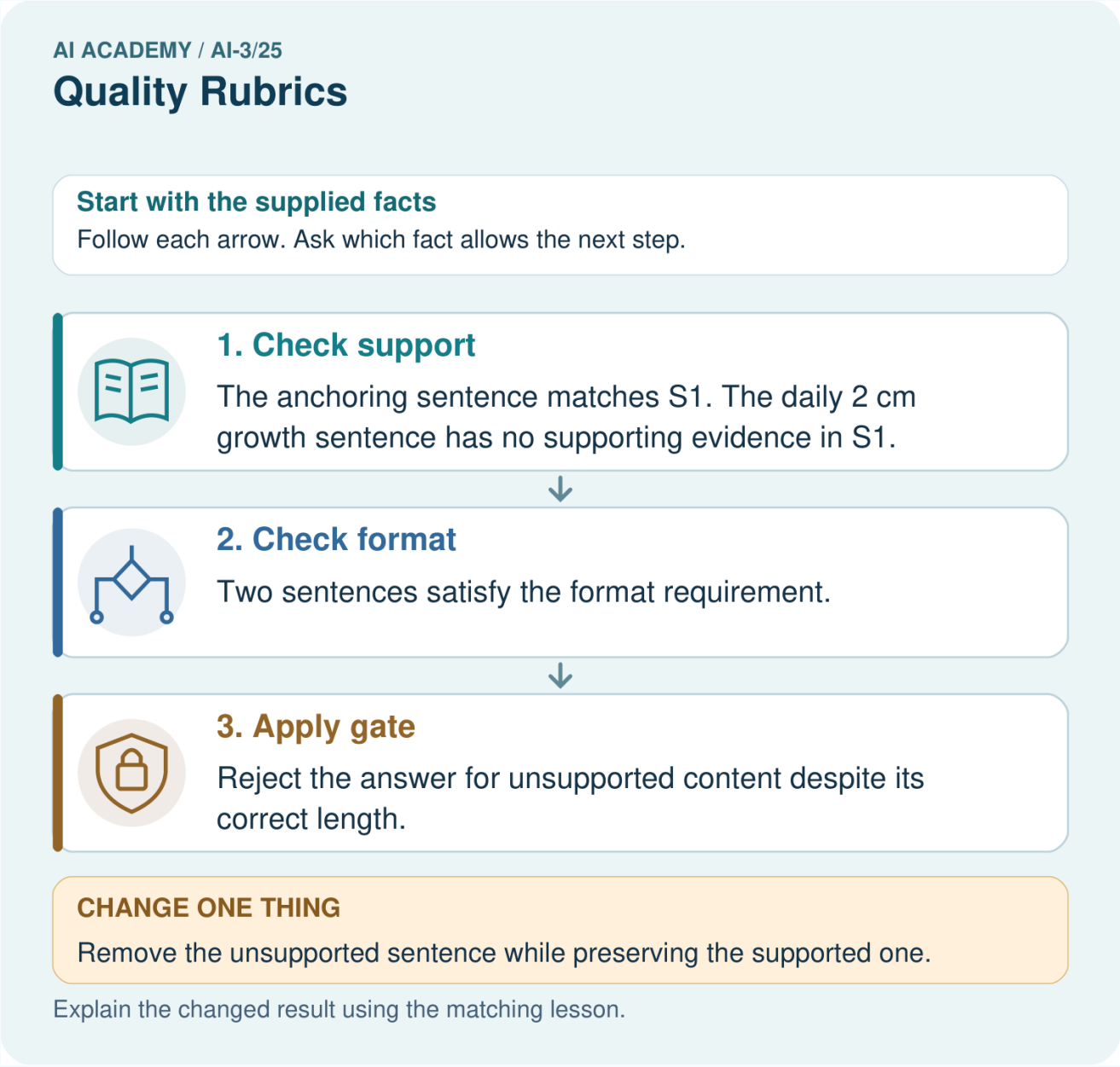
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Quality Rubrics

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide evidence-based rating, then remove help

> Collect individual judgements before negotiating disagreements.

Current completion task and key

Use S1: A root anchors a plant in soil. Question: what role does a root have? Draft: A root anchors a plant in soil. Roots grow exactly 2 cm each day. Apply source support, task fit and at-most-two-sentence format separately, giving evidence for each. Apply the declared all-three release rule, then predict removing only the unsupported sentence.

Support fails: daily growth has no S1 evidence. Task fit passes: the anchoring clause answers the role question. Format passes: two sentences. Do not release because support is mandatory and all three are required. Removing the growth sentence leaves a supported one-sentence role answer; all three then pass. No general readability or learning benefit is established.

Current individual gate check

A draft passes task fit and length but contains an unsupported measurement. Can a two-of-three score justify release under the stated all-three rule? Explain what must change.

No. Support is mandatory and all three checks must pass. Repair the unsupported claim, then recheck the whole output.

Pictorial changed-case reasoning

Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Support is a mandatory gate. Passing length cannot compensate for presenting an unsupported claim as fact. Changed-case check: The revised one-sentence answer passes these two declared checks.

The remaining claim is supported and one sentence is within the maximum.

Changed-case reasoning: The revised one-sentence answer passes these two declared checks. The remaining claim is supported and one sentence is within the maximum.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Rubric gives 0–2 each for support, usefulness and safety. Response is useful and safe but unsupported.

Write the question to explain: Why can a total mislead?

Reveal after **prediction**s: It may get 4/6 while failing essential support. Declare mandatory criteria before testing so other points cannot compensate.

Use the earlier changed question as optional later practice. Reserve Judge the new seed-coat draft independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

Qualification: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion task: Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for System evaluation sets. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the three-output rubric comparison

> Keep rating disagreements and source evidence visible.

Current cards and answer guidance

Three root cards: A support fails, task/format pass; B all three pass; C support/task pass, three-sentence format fails. Release only B under this small contract. Missing support is not proof of universal falsity. Independent seed-coat draft fails support for 48 hours; role and format pass, so reject and remove unsupported timing. Broad safety, readability and benefit remain unmeasured.

Earlier evaluation-rate tasks remain related review.

Predict, rate and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the three supplied root-role drafts and S1 with the declared source-support, task-fit and sentence-limit **anchor**s. Rate independently, justify each **dimension** and preserve the mandatory **evidence** gate. Revise ambiguous **anchor**s explicitly and rescore all **output**s; do not claim a live model or usability study.

4. What did the hardest test teach you that the normal test did not?

Keep evidence and limits

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Respond to rubric misunderstandings

Current misconceptions: high total permits a failed mandatory gate—use the declared release rule; unsupported means false—attribute the evidence gap; one rater's impression is an anchor—require a visible feature or source passage; changing an anchor improves measured results automatically—version and rescore all cases; three flags measure every quality—name the unmeasured dimensions.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Judge the new seed-coat draft independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new seed-coat rubric judgement

> Reveal this key only after the first attempt.

Current independent question and key

Approved C2 says: A seed coat protects the seed. Question: what role does the seed coat have? Draft: A seed coat protects the seed. Every seed opens exactly 48 hours later. Use the same three criteria: every factual claim supported by C2, answer the asked role question, at most two sentences. All three are required for release.

Give a separate judgement with evidence for each, the release decision and one justified repair.

Support fails: the 48-hour claim is absent from C2. Task fit passes through the protection clause; format passes with two sentences. Do not release under the all-three rule. Removing the unsupported timing sentence leaves the supported one-sentence role answer; do not invent a source or assert the timing claim is always false.

Judge reasoning and scope

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each **dimension** as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Current workbook guidance

Current W1 requires observable source, role and length anchors plus the all-three rule. W2 flags support fail/task pass/format pass, then all pass after removing unsupported growth. W3 rejects score compensation. W5 supplies a complete contrasting draft and versions an anchor revision before rescoring. The selected seed-coat task is independent. Earlier rate questions and keys remain review.

▼ Review · Earlier evaluation questions and keys

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A system evaluation set contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

W2 Trace the worked case

Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

W3 Audit the limit

A report omits this qualification: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Twelve cases: eight answerable, four unanswerable. Correct answers six; appropriate abstentions two. Compute separate and combined pass rates.

Teacher explanation: 6/8=75%; 2/4=50%; combined 8/12=2/3. Do not call all twelve answer-accuracy cases.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Selected case: Twelve cases: eight answerable, four unanswerable. Correct answers six; appropriate abstentions two. Compute separate and combined pass rates.

Selected case answer

Teacher explanation: 6/8=75%; 2/4=50%; combined 8/12=2/3. Do not call all twelve answer-accuracy cases.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

It may get 4/6 while failing essential support. Declare mandatory criteria before testing so other points cannot compensate.

Additional transfer practice question

Two answers have equally polished language. One has three supported claims; the other has two supported claims and one invented claim. How should a quality rubric distinguish them?

Reasoned practice answer

Score factual support separately from style and flag the invented claim. A serious unsupported claim can fail a required **evidence** criterion.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A single overall impression is enough to evaluate answer quality.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

No. Require source support as a mandatory pass condition in addition to any overall threshold. Keep tone and format as separate **dimension**s.

State the rule before testing to avoid approving unsupported answers because they look polished.

Explanations for the additional workbook examples

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Use a five-**dimension** rubric and block release when correctness or safety is below threshold.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Current independent task: seed-coat protection is supported by C2; 48-hour opening is not. Support fails, role and two-sentence format pass; reject under the all-three rule. Remove unsupported timing. Fresh retry: approved note says A shell protects an egg; draft adds exact three-day hatching. Require the same source-support separation without assuming a universal hatching time. Earlier rate calculations remain review; do not use their answers to score the rubric task.

Plan the delayed check

Suggested return points in this level: ai-3/26, ai-3/28, ai-3/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 26: Evaluation Sets. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 26: Evaluation Sets. Carry forward versioned criteria, mandatory conditions and justified case-level decisions when selecting cases and aggregating outcomes.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of system evaluation sets

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Practice and assessment reminders

Before class, solve the supported practice: Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

Check every learner before discussion: Use the worked case for System evaluation sets. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Remove the unsupported sentence while preserving the supported one. Require a **prediction** and a reason before comparing. Expected result: The revised one-sentence answer passes these two declared checks.

Reason: The remaining claim is supported and one sentence is within the maximum.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

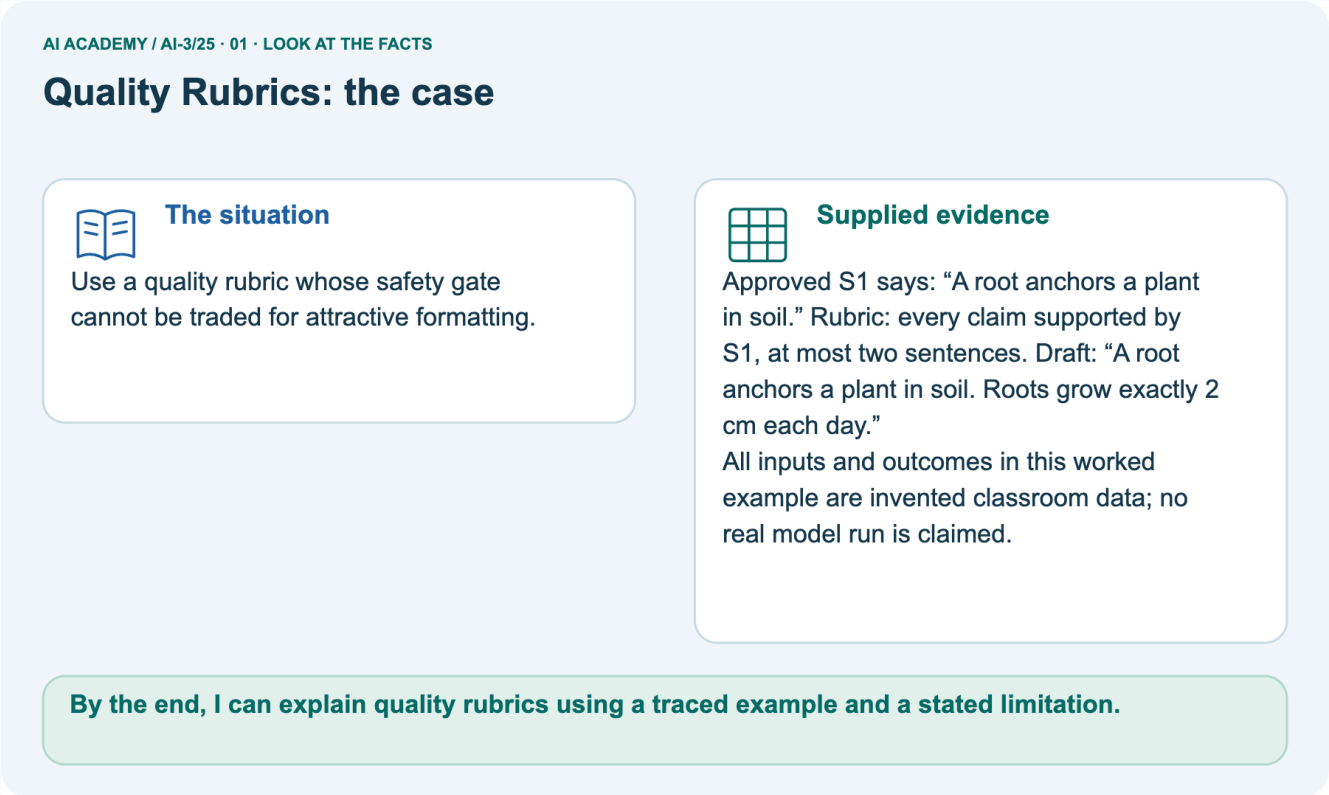
<details>
<summary>Reference: teach with the six pictures</summary>
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REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Quality Rubrics: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

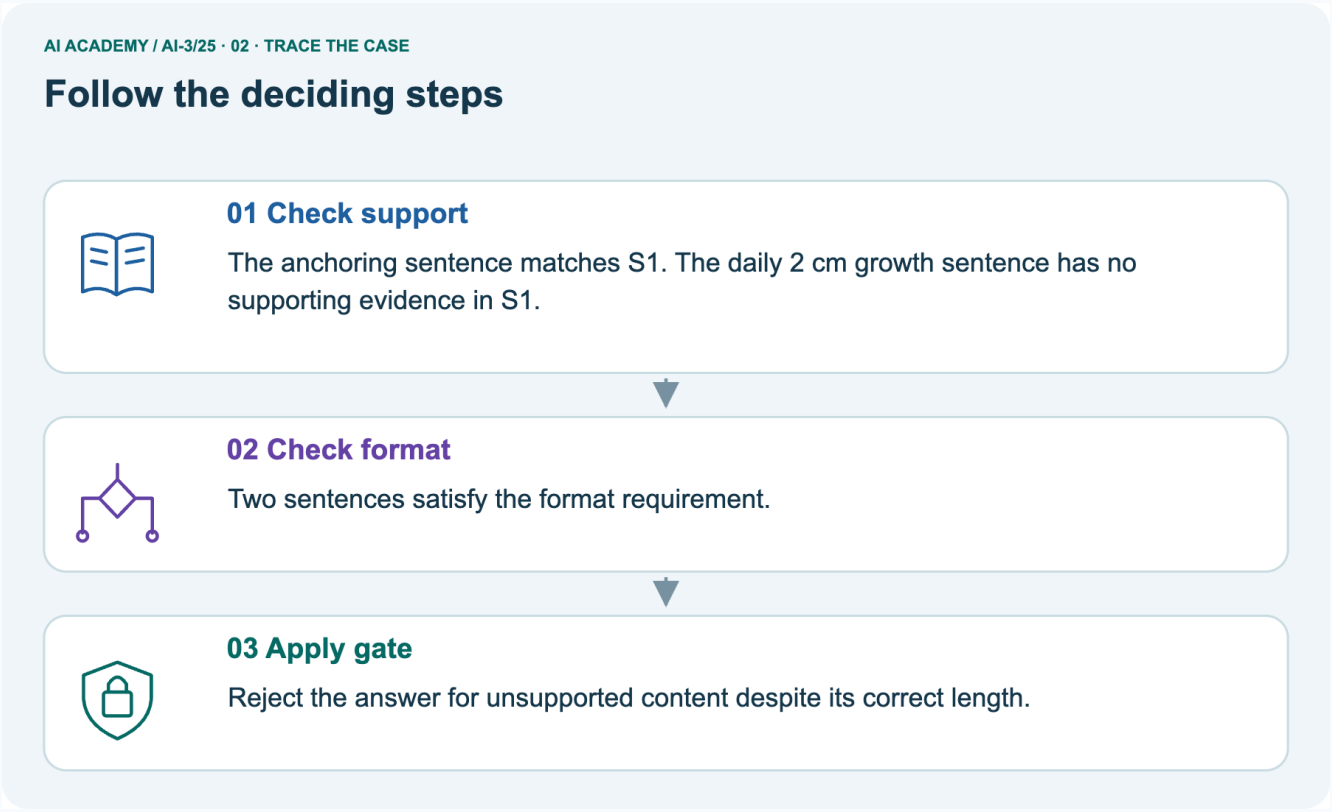
Use a quality rubric whose safety gate cannot be traded for attractive formatting. Supplied facts: Approved S1 says: “A root **anchor**s a plant in soil.” Rubric: every claim supported by S1, at most two sentences.

Draft: “A root **anchor**s a plant in soil. Roots grow exactly 2 cm each day.” All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

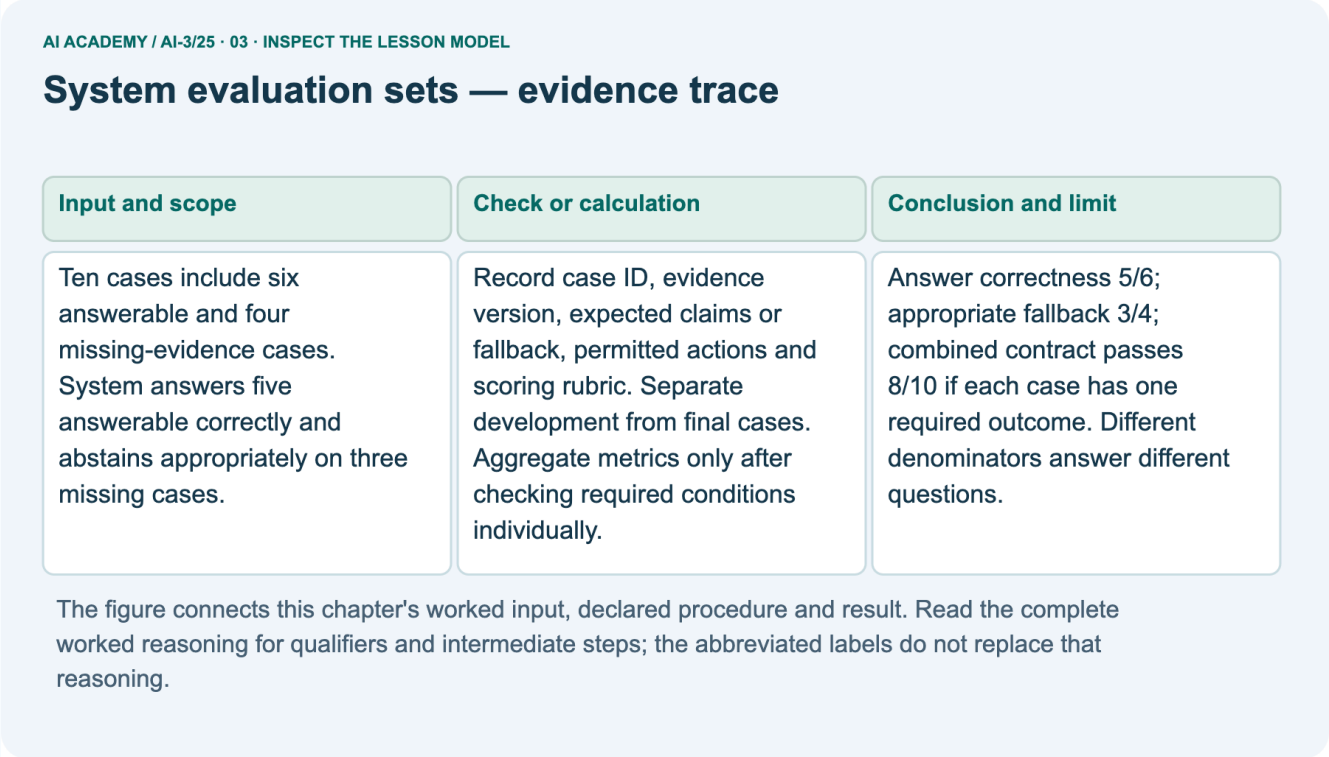
Check support: The **anchor**ing sentence matches S1.

The daily 2 cm growth sentence has no supporting **evidence** in S1. → Check format: Two sentences satisfy the format requirement. → Apply gate: Reject the answer for unsupported content despite its correct length.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. System evaluation sets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

System evaluation sets — **evidence** trace. Ten cases include six answerable and four missing-**evidence** cases.

System answers five answerable correctly and abstains appropriately on three missing cases. | Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric.

Separate development from final cases.

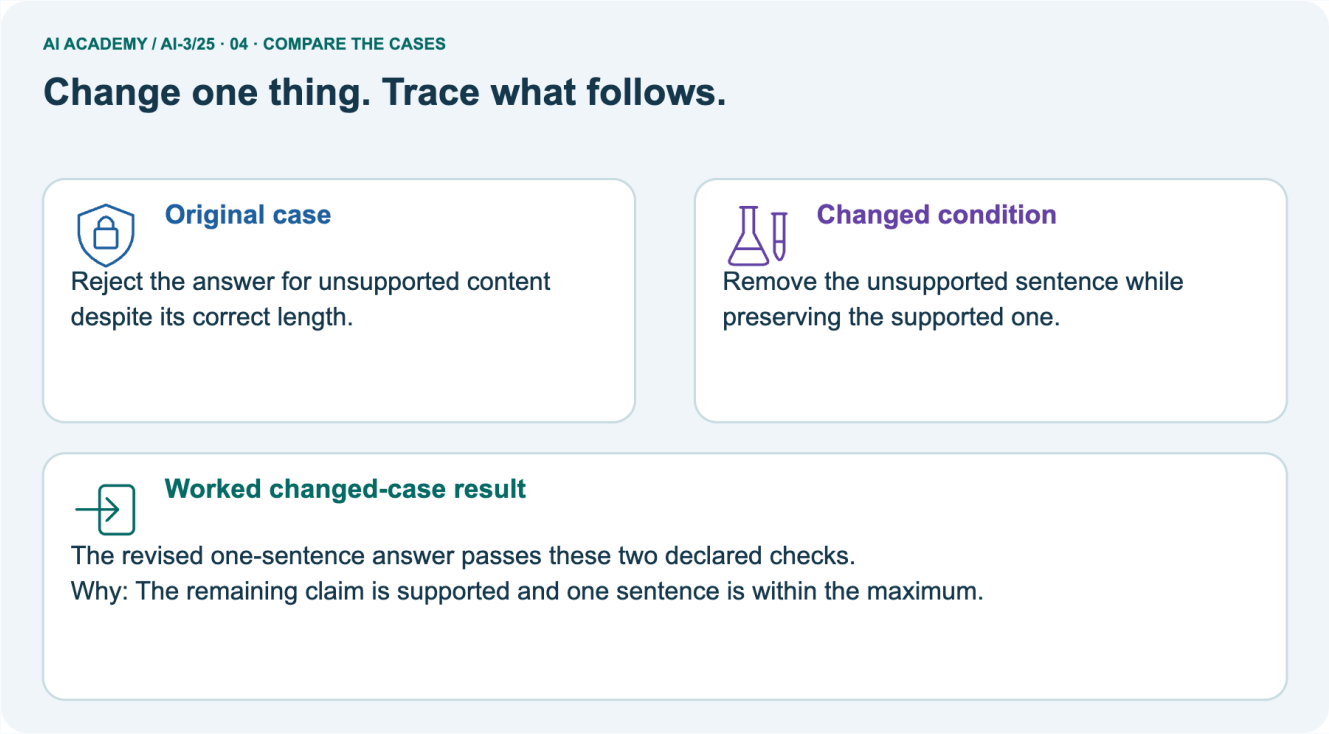
Aggregate metrics only after checking required conditions individually. | Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome.

Different denominators answer different questions.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Reject the answer for unsupported content despite its correct length. Changed condition: Remove the unsupported sentence while preserving the supported one.

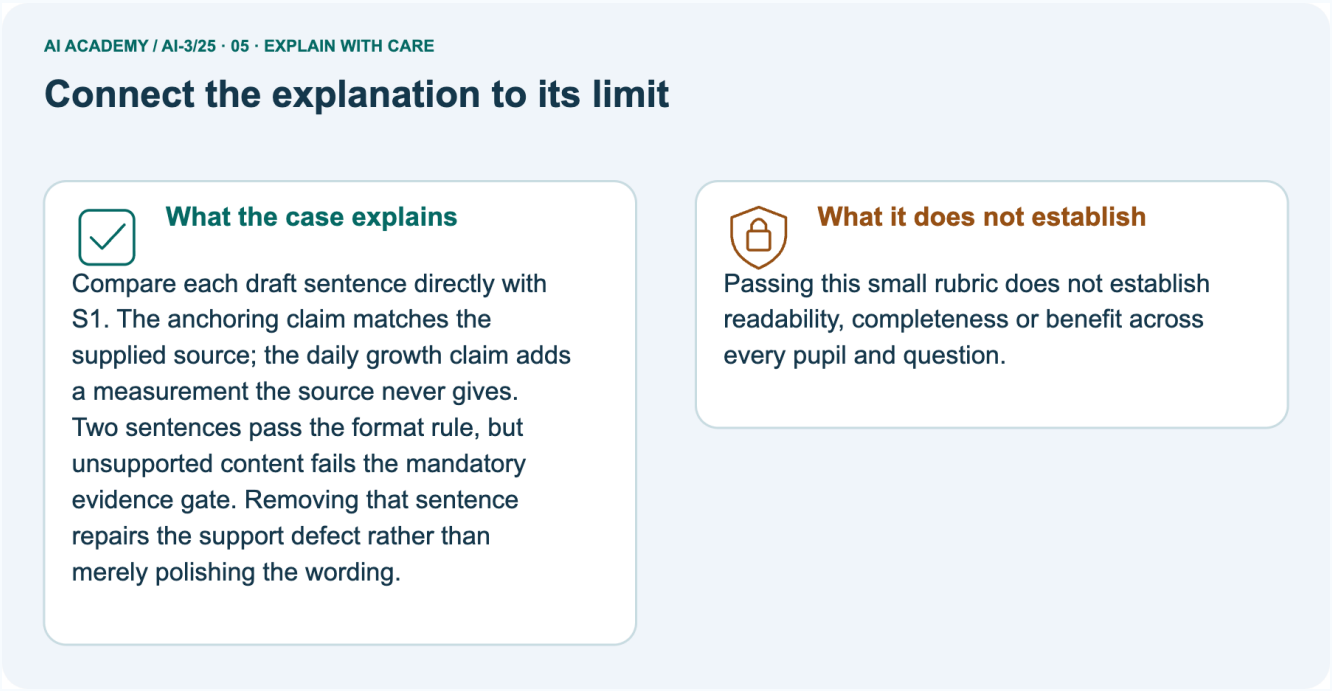
Expected result: The revised one-sentence answer passes these two declared checks. Reason: The remaining claim is supported and one sentence is within the maximum.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The remaining claim is supported and one sentence is within the maximum.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare each draft sentence directly with S1. The **anchor**ing claim matches the supplied source; the daily growth claim adds a measurement the source never gives.

Two sentences pass the format rule, but unsupported content fails the mandatory **evidence** gate. Removing that sentence repairs the support defect rather than merely polishing the wording.

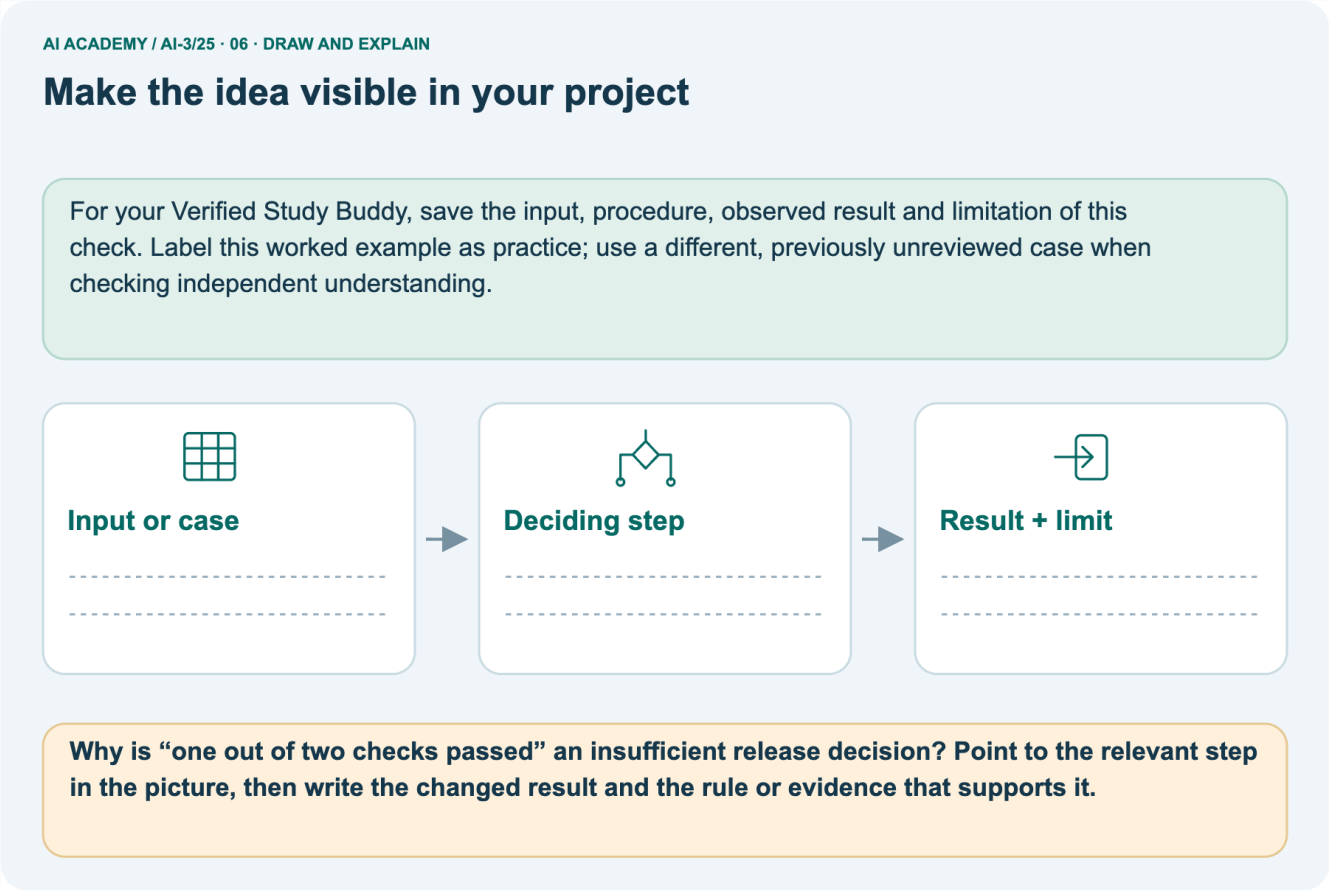
Limit: Passing this small rubric does not establish readability, completeness or benefit across every pupil and question.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Passing this small rubric does not establish readability, completeness or benefit across every pupil and question.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Support is a mandatory gate. Passing length cannot compensate for presenting an unsupported claim as fact. Changed-case check: The revised one-sentence answer passes these two declared checks.

The remaining claim is supported and one sentence is within the maximum.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Quality Rubrics: the case

02. Follow the deciding steps

03. System evaluation sets — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.